

MATHEMATICAL FOUNDATIONS OF POLITICAL METHODOLOGY: PLSC 43401

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Course description

This is a first course on the theory and practice of mathematical methods in social science research. These mathematical and computer skills are needed for the quantitative and formal modeling courses offered in the political science department, and are increasingly necessary for courses in American Politics, Comparative Politics, and International Relations. We will cover mathematical techniques (linear algebra, calculus, probability) and methods of logical and statistical inference (proofs and statistics).

Grading: You will be graded on 10 weekly problem sets (worth 40% of your total grade), occasional quizzes covering the reading (10%), and two, noncumulative exams (each worth 25% of your grade).

Problem sets are due each Tuesday at midnight, starting October 8th. Instructions for submission are as follows

- Each student, ideally working with a group, works through the problem sets without looking at the answers and writes up their first attempt (can be handwritten)
- Each student will, again as a group, discuss the correct answers and check.
- Using the correct answers, students will correct their written responses from step 1, explaining their errors, what they were missing, or describing what they still do not understand about the solution (this is written in Latex).

Grades are evaluated on the basis of clarity more than how correct they managed to get. The grader is your reviewer.

Required books: *Mathematics for Economists*, 4th Edition, by Pemberton & Rau, and *Introduction to Probability*, 2nd Edition, by Bertsekas & Tsitsiklis.

OCT 1: ALGEBRA, SETS, AND FUNCTIONS

- Pemberton & Rau Ch. 1-4

OCT 3: MATHEMATICAL AND SOCIAL SCIENTIFIC REASONING

- *Propositions and Patterns of Proof*
- Carroll, Lewis. "What the tortoise said to Achilles." *Mind* 104.416 (1995): 691-693.

OCT 8: SEQUENCES, SERIES AND LIMITS

- Pemberton & Rau Ch. 5, and skim Ch. 31

OCT 10: DIFFERENTIATION

- Pemberton & Rau Ch. 6, 7, appendix of Chapter 10

OCT 15: OPTIMIZATION

- Pemberton & Rau Ch. 8, 10

OCT 17: INTEGRATION

- Pemberton & Rau Ch. 19, 20

OCT 22: LINEAR ALGEBRA

- Pemberton & Rau Ch. 11, 12, 13.1

OCT 24: MULTIVARIATE CALCULUS

- Pemberton & Rau Ch. 13, 14, 15, 16

OCT 29: IN-CLASS MIDTERM EXAM

OCT 30: LAWS AND PROPERTIES OF PROBABILITY

- Bertsekas-Tsitsiklis Ch. 1.1-1.2

NOV 5: CONDITIONAL PROBABILITY

- Bertsekas-Tsitsiklis Ch. 1.3-1.7

NOV 7: DISCRETE RANDOM VARIABLES

- Bertsekas-Tsitsiklis Ch. 2

NOV 12: GENERAL (CONTINUOUS) RANDOM VARIABLES

- Bertsekas-Tsitsiklis Ch. 3

NOV 14: SUMMARIZING RANDOM VARIABLES

- Bertsekas-Tsitsiklis Ch. 3.4-3.7

NOV 19: COVARIANCE AND CORRELATION

- Bertsekas-Tsitsiklis Ch. 4.1-4.3

NOV 21: TRANSFORMATIONS OF RANDOM VARIABLES

- Bertsekas-Tsitsiklis Ch. 4.4-4.4.6

NOV 26: INEQUALITIES AND LIMITS OF SEQUENCES OF RANDOM VARIABLES

- Bertsekas-Tsitsiklis Ch. 5

NOV 28: HYPOTHESIS TESTING

- Bertsekas-Tsitsiklis Ch. 9
- Degroot and Schervish (optional) Chapter 8

DEC 3: CAUSAL INFERENCE

- Angrist Pischke *Mostly Harmless Econometrics* Sections 1, 2, 3.1 - 3.2 (pp 27 - 67), 4.1 - 4.2 (pp 113-146), 4.6 (pp 188- 215)

DEC 5: REVIEW